From continued fractions to certified spectra: the Gauss map, 200 years later

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A seminar for the Lindelöf Math Society

Plan of the talk

1. **Continued fractions** — an old friend from number theory.
2. **The Gauss map** — how the digits arise from a dynamical system.
3. **Gauss's question (1812)** — how are the digits distributed?
4. **Transfer operators** — turning a question about orbits into a question about a linear operator.
5. **A spectral theory primer** — eigenvalues of operators on infinite-dimensional spaces.
6. **A rigorous computation** — 50 eigenvalues, 90 digits each, certified by a computer.

No prior functional analysis required.

Continued fractions: examples

Every real number $x \in (0,1)$ has a **continued fraction expansion**

$$x = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}} = [a_1, a_2, a_3, \ldots], \qquad a_i \in \mathbb{N}.$$

$$\begin{aligned}
\tfrac{1}{2} &= [2] \\
\sqrt{2} - 1 &= [2, 2, 2, 2, \ldots] \text{ (purely periodic)} \\
\tfrac{\sqrt{5}-1}{2} &= [1, 1, 1, 1, \ldots] \text{ (golden ratio)} \\
e - 2 &= [1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \ldots] \text{ (has a pattern!)} \\
\pi - 3 &= [7, 15, 1, 292, 1, 1, 1, 2, 1, 3, \ldots] \text{ (no known pattern)}
\end{aligned}$$

- The digits a_i are integers, but their sequence encodes deep arithmetic.
- Truncating gives the **best rational approximations** of x — better than any other rational with the same denominator.

Why continued fractions matter

Theorem (best approximation)

If p/q is a convergent of x, then no rational p'/q' with $q' \leq q$ is strictly closer to x.

Example: $\pi = [3; 7, 15, 1, 292, \ldots]$.
The convergents are

$$3, \quad \frac{22}{7}, \quad \frac{333}{106}, \quad \frac{355}{113}, \quad \frac{103993}{33102}, \ldots$$

The enormous next digit 292 explains why $\frac{355}{113}$ is *astonishingly* good: $|\pi - 355/113| < 3 \cdot 10^{-7}$.

Big digits $\Leftrightarrow$ unusually good rational approximations.

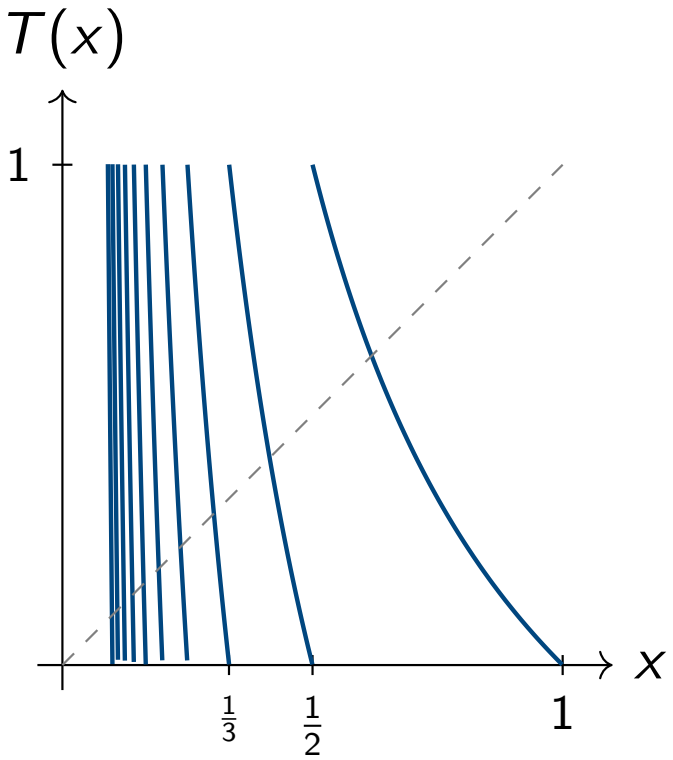
Natural questions:

- For a "typical" real x, how often is $a_i = 1$? $a_i = 2$? $a_i = 1000$?
- Is there a probability distribution on the digits?

The Gauss map

The digits a_i come from iterating a very simple function:

$$T : (0,1] \to [0,1), \qquad T(x) = \left\{\frac{1}{x}\right\} = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor.$$



Each branch is a hyperbola; as we move left, the branches get squeezed.

Iterating the Gauss map = reading off digits

Observation

If $x = [a_1, a_2, a_3, \ldots]$, then

$$a_1 = \lfloor 1/x \rfloor, \qquad T(x) = [a_2, a_3, a_4, \ldots].$$

So **iterating T shifts the digits to the left**.

Example.

$$x_0 = \pi - 3 \approx 0.14159\ldots \quad \Rightarrow \quad 1/x_0 = 7.0625\ldots, \quad a_1 = 7.$$

$$x_1 = T(x_0) = 0.0625\ldots, \quad 1/x_1 = 15.9966\ldots, \quad a_2 = 15.$$

$$x_2 = T(x_1) = 0.9966\ldots, \quad 1/x_2 = 1.0034\ldots, \quad a_3 = 1.$$

And so on: $\pi - 3 = [7, 15, 1, 292, \ldots]$.

The orbit $(x_0, x_1, x_2, \ldots)$ is the continued fraction of x_0.

Gauss's question (1812)

"Let x_0 be uniformly random in $(0,1)$. What is the distribution of $x_n = T^n(x_0)$?"

— Gauss to Laplace, 1812

Let $F_n(x) = \mathbb{P}(x_n \le x)$. Gauss conjectured

$$F_n(x) \xrightarrow[n\to\infty]{} \log_2(1+x).$$

This $\log_2(1+x)$ is the **Gauss measure**. Its density is

$$\rho(x) = \frac{1}{\ln 2} \cdot \frac{1}{1+x}.$$

Consequences (Gauss–Kuzmin, 1928):

- $\mathbb{P}(a_i = k) = \log_2\left(1 + \frac{1}{k(k+2)}\right)$.
- $\mathbb{P}(a_i = 1) \approx 41.5\%$.
- $\mathbb{P}(a_i = 2) \approx 17.0\%$.
- The digits of *almost every* real are distributed this way.

The rate of convergence

Gauss proved $F_n \to \log_2(1+x)$ but **not** the rate.

Lévy (1929), Kuz'min: convergence is exponential,

$$\sup_x |F_n(x) - \log_2(1+x)| \;\le\; C \cdot q^n$$

for some $0 < q < 1$.

Wirsing (1974): identified the sharp rate as

$$q = |\lambda_2| = 0.30366\,30029\ldots$$

where λ_2 is the **second eigenvalue** of a certain linear operator. Wirsing computed 10 digits by an iterative procedure; Babenko (1978), Flajolet–Vallée (1995), Sebah, and Briggs (2003) progressively pushed the precision much further.

A question about orbits of T became a question about **eigenvalues**.

How did that happen?

From orbits to densities: the transfer operator

If $x_0 \sim \rho_0$ (density on $(0,1]$), what is the density of $x_1 = T(x_0)$?

Change of variables for each branch $T_k^{-1}(y) = \frac{1}{y+k}$:

$$\rho_1(y) = \sum_{k\geq 1} \rho_0\left(\frac{1}{y+k}\right) \cdot \left|\frac{d}{dy}\frac{1}{y+k}\right| = \sum_{k\geq 1} \frac{1}{(y+k)^2}\,\rho_0\left(\frac{1}{y+k}\right).$$

The Gauss–Kuzmin–Wirsing (GKW) operator

$$(\mathcal{L}f)(x) \;=\; \sum_{k=1}^{\infty} \frac{1}{(x+k)^2}\, f\left(\frac{1}{x+k}\right).$$

$\mathcal{L}$ is **linear**: $\mathcal{L}(\alpha f + \beta g) = \alpha \mathcal{L} f + \beta \mathcal{L} g$.

And it **iterates**: the density of x_n is $\mathcal{L}^n \rho_0$.

What the GKW operator tells us

Fixed point: $\mathcal{L}\rho = \rho$ where $\rho(x) = \frac{1}{\ln 2} \cdot \frac{1}{1+x}$.
This is the Gauss invariant density — the "equilibrium" of the dynamics.

Spectral expansion (informal): if $\mathcal{L}$ has eigenvalues $1 = \lambda_1 > |\lambda_2| \geq |\lambda_3| \geq \cdots$ with eigenfunctions v_j and dual functionals ℓ_j, then

$$\mathcal{L}^n f \;\approx\; \underbrace{\rho \cdot \int_0^1 f}_{\text{equilibrium}} \;+ \lambda_2^n\, \ell_2(f)\, v_2 \;+\; \lambda_3^n\, \ell_3(f)\, v_3 \;+\; \cdots$$

Each eigenvalue λ_j contributes a mode that decays at rate $|\lambda_j|^n$.

The biggest non-trivial eigenvalue governs the slowest decay: that is why Gauss's convergence rate is $|\lambda_2|^n$.

Computing many eigenvalues of $\mathcal{L}$ = understanding the dynamics in detail.

A short primer on spectra

Finite dimensions. If $A \in \mathbb{C}^{n \times n}$, the spectrum is

$$\sigma(A) = \{\lambda \in \mathbb{C} : \det(\lambda I - A) = 0\} = \{\text{eigenvalues}\}.$$

Finite set. Easy.

Infinite dimensions. For a bounded linear operator $L : \mathcal{B} \to \mathcal{B}$ on a Banach space, the spectrum is

$$\sigma(L) = \{\lambda \in \mathbb{C} : \lambda I - L \text{ is not invertible}\}.$$

- It can be a very nasty subset of $\mathbb{C}$ — an arc, a disc, even a fractal.
- Eigenvalues are only *part* of the spectrum.
- The **resolvent** $(zI - L)^{-1}$ exists for $z \notin \sigma(L)$ — and blows up as z approaches $\sigma(L)$.

Good news: for **compact** operators, the non-zero part of the spectrum consists only of isolated eigenvalues of finite multiplicity — it looks just like the matrix case.

Compact operators: the nice case

Riesz–Schauder theorem (informal)

If $L : \mathcal{B} \to \mathcal{B}$ is compact, then

$$\sigma(L) \;=\; \{0\} \cup \{\lambda_1, \lambda_2, \lambda_3, \ldots\}$$

where the λ_j are eigenvalues with $\lambda_j \to 0$.

Example. Integral operators $(Kf)(x) = \int_0^1 k(x,y) f(y)\, dy$ with smooth kernel.

The GKW miracle: on the right function space, the Gauss–Kuzmin–Wirsing operator $\mathcal{L}$ is **compact**.

So:

- It has a sequence of eigenvalues $\lambda_j \to 0$.
- They are the only obstruction to $\mathcal{L}^n f$ converging to equilibrium.
- Understanding $\mathcal{L}$ = computing these numbers.

The right function space: $H^2(D_1)$

Which space makes $\mathcal{L}$ compact? We use a **Hardy space** of holomorphic functions.

Setup: let $D_1 = \{w \in \mathbb{C} : |w-1| < 1\}$ — an open disc containing the interval $(0, 1]$.

$$H^2(D_1) = \left\{ f : D_1 \to \mathbb{C} \text{ holomorphic}, \ \|f\|_{H^2}^2 := \sup_{r<1} \int |f|^2 < \infty \right\}.$$

Why this works: the inverse branches $\tau_k(w) = \frac{1}{w+k}$ map a *larger* disc $D_{3/2}$ *into* D_1.

The key geometric fact ("domain gain")

$\tau_k(D_{3/2}) \subset D_1$ for every $k \geq 1$.

So $\mathcal{L}$ sends $H^2(D_1)$ into the smaller space $H^2(D_{3/2})$, and the inclusion $H^2(D_{3/2}) \hookrightarrow H^2(D_1)$ is compact.
This is what makes $\mathcal{L}$ compact.

The Gauss–Babenko–Knuth problem

By the 1970s it was understood that the GKW constant $|\lambda_2|$ is an eigenvalue of $\mathcal{L}$.

Babenko (1978): computed λ_2 to about 40 decimal digits numerically.

Knuth, The Art of Computer Programming (3rd ed., 1997):

> *"Design a method for computing λ_j and the eigenfunctions $\Psi_j(x)$ to high precision."*

This is the **Gauss–Babenko–Knuth problem**.

What was missing for 25+ years:

- **Rigorous** error bounds on *each* computed eigenvalue (not just λ_2).
- A certified method for **many** eigenvalues at once.
- Control of eigenvectors and spectral projectors, not just eigenvalues.

Why "certified" is harder than it looks

Floating-point computation lies. A numerical routine can return "$\lambda_2 \approx -0.303663\ldots$" with no guarantee of correctness — rounding errors accumulate, and you cannot tell *how many* of the digits are right.

Two deeper problems:

1. **Discretization error.** We approximate the infinite-dimensional $\mathcal{L}$ by a finite matrix L_K of size $(K+1) \times (K+1)$. How close is $\sigma(L_K)$ to $\sigma(\mathcal{L})$?
2. **Spectral pollution.** Eigenvalues of L_K may converge to points *not* in $\sigma(\mathcal{L})$, or miss points that *are*.

Colbrook–Hansen (2023, SCI hierarchy): for a *general* bounded operator given only by its matrix entries, there is **no algorithm** that decides, at any finite step, whether a given complex number is in the spectrum.

So how can we hope to get 90 digits right?

Main result

Theorem (Nisoli, 2026)

The first 50 nonzero eigenvalues of the GKW operator on $H^2(D_1)$ are real and simple. Each is known with a **certified enclosure of at least 90 decimal digits**.

In particular,

$$\tilde{\lambda}_2 = -0.30366\,30028\,98732\,65859\,74481\ \ldots$$

(160 digits shown, 90 of them rigorously certified).

Also certified:

- the eigenfunctions Ψ_j (as elements of $H^2(D_1)$),
- the **Riesz spectral projectors** onto each eigenspace,
- the **spectral gap**: a quantitative enclosure of the annulus separating λ_j from the rest.

How much computing? A single desktop workstation.

The central idea: spectral gap as *output*

Old paradigm. Assume a spectral gap exists $\Rightarrow$ derive quantitative consequences.

New paradigm (this work).

Compute finite-rank data $\Rightarrow$ **certify** spectral gap *a posteriori*.

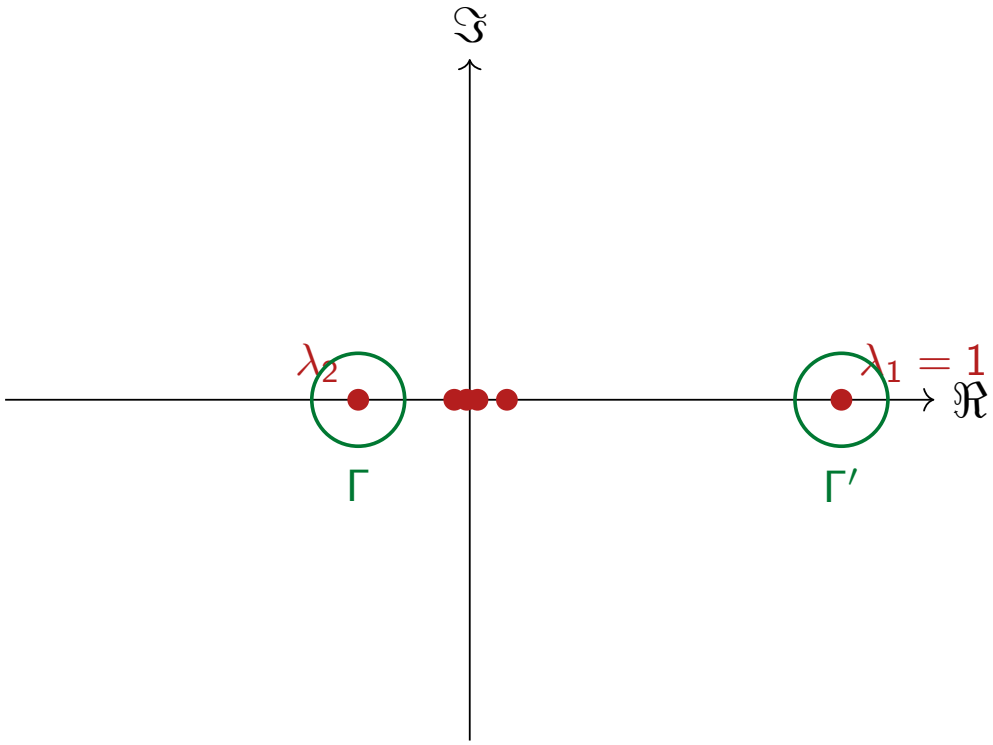
The slogan

A single inequality, checked at the end of the computation, simultaneously certifies:

- that the true spectrum is where the computed one says it is,
- that no eigenvalues were missed,
- that the spectral gap has the claimed size,
- that eigenvectors and projectors are close to the computed ones.

The inequality involves a **resolvent bound along a contour**.

A picture of the idea



Pick a contour $\Gamma \subset \mathbb{C}$ around one (or several) eigenvalues you want.

If you can bound the resolvent $\|(zI - L_K)^{-1}\|$ along Γ *computationally*, and the bound is good enough, then:

- Γ lies in the resolvent set of the true $\mathcal{L}$,
- inside Γ, $\mathcal{L}$ and L_K have the same eigenvalues (counted with multiplicity).

The three inputs

Given a compact operator $\mathcal{L} : \mathcal{B} \to \mathcal{B}$:

(A) A finite-rank truncation L_K with explicit error bound

$$\|\mathcal{L} - L_K\| \leq \varepsilon_K.$$

(B) A certified bound on the resolvent of L_K along a contour Γ,

$$M_\Gamma := \sup_{z \in \Gamma} \|(zI - L_K)^{-1}\| < \infty.$$

(C) The **key inequality**

$$\varepsilon_K \cdot M_\Gamma < 1.$$

Conclusion

When (A), (B), (C) hold: $\Gamma \subset \rho(\mathcal{L})$, and $\mathcal{L}$ and L_K have the **same eigenvalues inside** Γ with the same multiplicities.

Proof idea. Write $zI - \mathcal{L} = (I - (\mathcal{L} - L_K)(zI - L_K)^{-1})(zI - L_K)$ and invert the first factor by Neumann series. This is essentially the *resolvent identity* — one line.

Where do ε_K and M_Γ come from?

(A) The truncation bound ε_K: provided by the **domain gain** for the Gauss map.

The Taylor coefficients of $\mathcal{L}f$ around $w = 1$ decay **geometrically** by a factor $(2/3)^K$:

$$\|\mathcal{L} - L_K\|_{H^2(D_1)} \;\leq\; C \cdot \left(\tfrac{2}{3}\right)^{K+1}, \qquad C \approx 10.058.$$

At $K = 512$: $\varepsilon_K \approx 10^{-90}$.

(B) The resolvent bound M_Γ: computed from the *finite matrix* L_K

- Schur-decompose L_K into an almost-triangular form.
- At many sample points $z_\ell \in \Gamma$, use a certified smallest-singular-value routine to get $\|(z_\ell I - L_K)^{-1}\| \leq 1/s_\ell$.
- Propagate between samples by Lipschitz continuity of the resolvent.
- All arithmetic is done with **interval / ball arithmetic** (Arb library, Julia) so every rounding error is tracked.

The computational pipeline

Compute matrix entries via Hurwitz zeta (ball arithmetic)

↓

Certified Schur decomposition of L_K

↓

Sample contour Γ; certify $\sigma_{\min}$

↓

Check the key inequality $\varepsilon_K M_\Gamma < 1$

↓

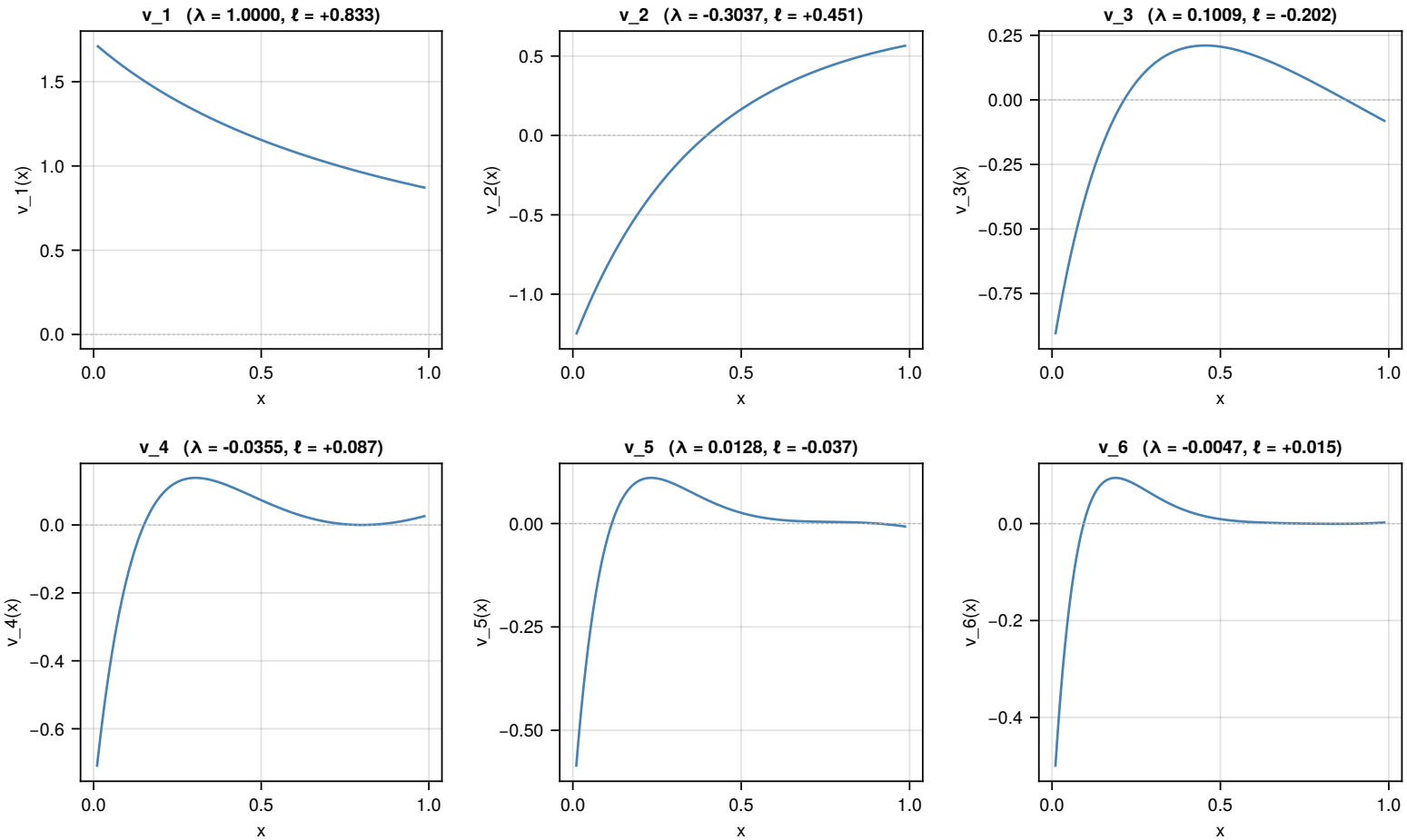
Riesz projectors $\Rightarrow$ eigen-vectors and eigenvalues

↓

Output: certified digits for all 50 eigenvalues

$K = 1024$, 2048-bit arithmetic, single desktop workstation.

The certified eigenfunctions



First six certified eigenfunctions $\Psi_1, \ldots, \Psi_6$ of the GKW operator. Ψ_1 (top left) is the Gauss invariant density $\frac{1}{\ln 2} \cdot \frac{1}{1+x}$, positive everywhere. Higher eigenfunctions oscillate more, reflecting the finer structure of the continued-fraction dynamics.

Breaking the SCI barrier

Recall: for a *general* bounded operator, no algorithm certifies spectral data at finite steps (Colbrook–Hansen).

Why we escape

The impossibility assumes nothing but matrix entries. But transfer operators come with **dynamical structure**:

- the *domain gain* gives a computable bound ε_K,
- a *Lasota–Yorke inequality* controls the functional-analytic side,
- the resolvent of the finite matrix is certifiable.

Together, these furnish the "oracle" missing from the abstract setting.

Dynamics is not just the problem — it is the way out.

Beyond the Gauss map

The three-input framework applies whenever one has:

- a compact transfer operator (with explicit truncation bounds),
- or an operator satisfying a Lasota–Yorke inequality (piecewise expanding maps, some hyperbolic maps).

Possible targets:

- Rigorous mixing rates for Markov processes with a Doeblin–Fortet inequality.
- Renormalization operators: certified hyperbolicity of Feigenbaum's period-doubling fixed point (Lanford 1982, systematic version).
- Fredholm determinants / dynamical zeta functions.
- Koopman operators with noise-induced compactness.

Current limitation. The method certifies *isolated* spectral data. It cannot say anything about continuous spectrum or eigenvalues buried inside the essential spectral radius.

Summary

What we did

1. Built an abstract framework that turns finite-rank data into certified spectral data for compact (or DFLY) operators — **no spectral gap assumed as input**.
2. Applied it to the Gauss–Kuzmin–Wirsing operator to solve the Gauss–Babenko–Knuth problem: **50 eigenvalues, each certified to $\geq$ 90 digits**, plus eigenfunctions, projectors and spectral gap.

Key messages for the Lindelöf Math Society:

- Dynamics $\to$ operators $\to$ linear algebra $\to$ rigorous digits.
- Interval arithmetic turns a computer into a proof-checker.
- The spectral gap is an *output*, certified a posteriori.
- Classical questions can still be settled 200 years later.

Thank you

Thank you!

Questions?

Code: `https://github.com/orkolorko/GKWExperiments.jl`

Data: `doi:10.7910/DVN/HKM3Y2`

Preprint: `https://arxiv.org/abs/2602.19435`

Suggested reading:

- Khinchin, *Continued Fractions* (a short classic).
- Einsiedler–Ward, *Ergodic Theory with a View Towards Number Theory*, Chapter 3.
- Baladi, *Positive Transfer Operators and Decay of Correlations*.
- Tucker, *Validated Numerics* (on interval arithmetic).